

Weekly Diary – Week 16

Internship Organization: Vikaspedia

Duration: 05/05/2026 – 10/05/2026

Objective of the Week

The primary objective of this week was to implement keyword demand validation, caching optimization, and frontend integration for displaying generated keyword results. The focus was on improving performance while validating keyword usefulness using trend and autocomplete data.

Date – 05 May

Google Autocomplete integration was implemented to validate whether generated keywords matched real user search behavior. Lightweight validation logic was added to reduce unnecessary external requests while maintaining keyword quality.

Date – 06 May

Redis caching was integrated into the keyword validation pipeline to store previously validated keywords and reduce repeated API processing. Cache expiration policies and retrieval logic were optimized for low-latency responses.

Date – 07 May

Support for trend-based validation using Pytrends data was added. The system was designed to estimate keyword popularity using cached trend information and assign metadata such as trend direction and validation status.

Date – 08 May

The React frontend interface was connected to the backend keyword generation API. Content input forms, API request handling, and keyword display components were implemented using React and Vite.

Date – 09 May

Frontend improvements were made to display keyword metadata such as keyword type, popularity indicators, validation source, processing time, and estimated search demand. UI responsiveness and layout adjustments were also completed.

Date – 10 May

End-to-end testing of frontend and backend integration was performed. Keyword generation, validation, caching, and API response handling were verified to ensure stable and accurate system functionality.